

Programming Techniques – Using C Language**Structures****Objective:**

The purpose of this lab is to get an idea about structures in c.

1. Type and run the given program to understand the usage of structures.

```
#include<stdio.h>
#include<string.h>

struct lib_books{
    char title[20];
    char author[15];
    int pages;
    float price;
};

main(){
    struct lib_books book1; // Line A
    strcpy(book1.title, "Introduction to C"); // Line B
    strcpy(book1.author, "Dennis"); // Line C
    book1.pages=100; // Line D
    book1.price=450.00; // Line E
    printf("Book Details");
    printf("\nTitle: %s\nAuthor: %s\nNo of Pages: %d\nPrice: %.2f\n", book1.title, book1.author, book1.pages,
    book1.price);
    return 0;
}
```

- a) Comment your **Line B,C,D,E** of above program and modify your Line A to initialize the book1 with values and execute it.
struct lib_books book1={"Introduction to C", "Stephan G.K.", 200, 560.00};
- b) Modify above program to read the book1's data from the keyboard and display it.
- c) Modify your program to handle the three different books of lib_books type.

2. Define a student structure as follows

```
struct student
{
    int id_no;
    char name[20];
    char address[20];
    char combination[5];
    int age;
}newstudent;
```

- a) Write a C program to handle the new student with suitable data.
- b) Relevant data must read from the keyboard.
- c) Output the newstudent data with the suitable headings.

3. Type and run the given program to understand the usage of structures with **typedef** keyword.

```
#include<stdio.h>
#include<string.h>
typedef struct lib_books {
    char title[20];
    char author[15];
    int pages;
    float price;
}lib_books;

int main(){
    lib_books book1={"Introduction to C","Arther",200,560.00};
    printf("Book Details");
    printf("\nTitle: %s\nAuthor: %s\nNo of Pages: %d\nPrice: %.2f\n",book1.title,book1.author,book1.pages,book1.price);
    return 0;
}
```

4. Modify above program to do the followings

- a) Declare an array to store the details of 5 books.
- b) Input the details of the books from the keyboard.
- c) Display the titles of the books, prices of which are below Rs. 2000.00.
- d) Suppose a customer is interested in buying books written by the author "Kernighan". Display the titles and the prices of the books written by the above author.

5. Define a student structure as follows.

```
typedef struct student_type
{
    char name[20];
    int ID;
} student_type;
```

- a. Write a C program to handle the new student of **student_type**.
- b. Relevant data must read from the keyboard.
- c. Output the newstudent data with the suitable headings.
- d. Modify your program to set the student data using a separate function and it should return the complete student data.
- e. Add another function which displays student data after passing the **student_type** variable.

6. Write a C program according to the given guidelines.
- Define a structure Date to store the date as day, month and year separately as integer values.
 - Define another structure Employee to store the employees' details (name of the employee, salary and the hired date). Hint: use nested structures. Hired date must be Date type.
 - Write a function called GetDate() that will receive no arguments and ask for the date in numerical format. Then return the date as struct Date.
 - Use the GetDate() function inside the main method and set the date to March 15,2000 in suitable format and display the date.
 - Write a function called PrintEmp() that must has a one argument of type struct Employee and it must print its EmpName, Salary and Date hired.
 - Create Employee type employee called Person1 and set the suitable data to Person1. Then print your Person1's data by calling your PrintEmp() function.
7. Type and execute following program and get an idea about structures and pointers.

```
#include <stdio.h>
#include <string.h>

struct Emp{
    char lname[20]; /* last name */
    char fname[20]; /* first name */
    int age;        /* age */
    int rate;       /* e.g. 12 per hour */
};

int main(){
    struct Emp employee1; /* declare the structure variable */
    struct Emp *pEmp;     /* declare the pointer to structure */
    pEmp=&employee1;
    strcpy(employee1.lname,"Kumara");
    strcpy(employee1.fname,"Saman");
    employee1.age=27;
    employee1.rate=13;
    printf("first name=%s\n",pEmp->fname);
    printf("last name=%s\n",pEmp->lname);
    printf("age=%d\n",pEmp->age);
    printf("rate=%d\n",pEmp->rate);
    return 0;
}
```